DAI VIET HUNG TRAN

AI ENGINEER | COMPUTER VISION ENGINEER

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PROFESSIONAL SUMMARY

Computer Science M.S. student at Sejong University with a focus on Artificial Intelligence and Computer Vision. Experience includes computer vision research, AI backend development, edge computing, and retrieval-augmented generation across academic and industry projects. Comfortable working across the AI development lifecycle, from data preparation and model training to evaluation, API integration, optimization, and deployment. Primary technical interests include applied computer vision, real-time AI systems, and practical machine learning solutions.

TECHNICAL SKILLS

Core: Python, Computer Vision, Deep Learning, Machine Learning, RAG, AI Backend Development
Computer Vision: PyTorch, SegFormer-B1, YOLOv8, OpenCV, Image Segmentation, Object Detection, InsightFace/ArcFace, ByteTrack
LLM / Retrieval: Gemma 3, nomic-embed-text-v2-moe, FAISS, LangChain, Embeddings, Recursive Character Text Splitting
ML / Optimization: TensorFlow, scikit-learn, Random Forest, Genetic Algorithms, Grey Wolf Optimization, KNIME, ComfyUI
Backend / Data: FastAPI, Flask, PostgreSQL, Supabase, MongoDB, SQL, NumPy, Pandas
Deployment / Programming: Docker, Vercel, Cloudflared, Git | C++, Java, C, Swift, JavaScript

PROFESSIONAL EXPERIENCE

Research Assistant | Sejong University **Aug 2026 - Present**
Seoul, South Korea

- Develop an **edge-AI pipeline on Jetson Orin Nano** for UAV-based coffee-flower density estimation and optimal beehive placement in agricultural fields.
- Built the end-to-end **SegFormer-B1** segmentation workflow at **640x640** input resolution using approximately **500 labeled images** across three flower-density classes; achieved **0.67 validation mIoU**.
- Implemented training, preprocessing, validation, inference, and model-serving components independently, with a **FastAPI** backend connecting computer-vision outputs to downstream optimization.
- Designed a **Genetic Algorithm** objective using flower density, water availability, weather, travel distance, and bee-activity features to recommend practical hive-placement locations.
- Own the full technical workflow from model experimentation to edge inference and backend integration, enabling rapid iteration without dependency on separate ML or backend teams.

Project Manager | EBIT **May 2026 - Aug 2026**
Seoul, South Korea

- Coordinated **600+ international contributors** with support from assistant leads across two concurrent AI data and benchmark programs conducted primarily in English.
- Managed a multilingual speech-data collection program for AI training and a terminal benchmark project evaluating **ChatGPT and Claude** on technical tasks.
- Worked directly with clients on requirements, delivery planning, quality issues, and project changes while translating technical constraints into actionable team workflows.
- Mentored junior members, reviewed work quality, and coordinated assistants to keep large distributed teams aligned across languages and time zones.

AI Engineer | EBIT **Dec 2025 - Aug 2026**
Seoul, South Korea

- Developed AI algorithms for an intelligent Korean military meal-planning platform used by approximately **1,000 active users**.
- Integrated **Gemma 3** with **nomic-embed-text-v2-moe** to interpret nutritional information and adapt recommendations to military activity profiles and meal constraints.
- Built a **RAG-based nutrition assistant** enabling officers to query and understand daily, weekly, and monthly nutritional plans using natural language.
- Integrated **Supabase/PostgreSQL** and optimized query patterns for responsive retrieval and data access; the system was demonstrated at **AI Expo 2026**.

Student Researcher, XR Research Laboratory | Sejong University **Mar 2025 - May 2026**
Seoul, South Korea

- Researched diffusion-model workflows with **ComfyUI and KNIME** and built reusable experimental pipelines for model generation, parameter exploration, and workflow optimization.
- Developed a **Random Forest regression** pipeline for earthquake-area prediction and applied **Grey Wolf Optimization** for model and parameter tuning.
- Conducted experiment design, model evaluation, and technical documentation across applied machine-learning and deep-learning research tasks.

Software Development Intern | Sejong University **Sep 2023 - Mar 2025**
Seoul, South Korea

- Developed the backend and retrieval pipeline for a four-person RAG research system processing **100,000+ pages** of Masnavi literature.
- Implemented **FAISS** retrieval with **nomic-embed-text-v2-moe** and Recursive Character Text Splitting; optimized chunking and vector indexing to improve **RAM usage, retrieval latency, and answer accuracy**.

AI Programmer | Outlier **Dec 2024 - Mar 2025**
Remote
- Evaluated and improved AI-generated Python solutions, reviewing technical correctness, reasoning quality, edge cases, and code reliability for model-training and benchmarking tasks.
- Compared model outputs against task requirements and corrected weak reasoning or implementation details to improve response quality and consistency.

SELECTED PROJECTS

Smart Aquaculture Assistant | *Python | YOLO | Computer Vision | Real-Time Inference*
- Built a real-time YOLO-based fish-disease detection pipeline and achieved **0.89 mAP** during model evaluation.
- Measured approximately **60 FPS** with about **10 ms model inference latency** on a MacBook M2 with 16 GB RAM.
- Integrated detection results into a broader smart-aquaculture assistant, connecting computer vision with user-facing AI assistance for disease monitoring and decision support.
- Received the **People's Choice Award** at Sejong University Capstone Design 2025.

Real-Time Missing Person Detection | *YOLOv8 | InsightFace/ArcFace | ByteTrack | Docker*
- Built a real-time identity-tracking pipeline combining YOLOv8 person detection, **512-d ArcFace embeddings**, cosine similarity matching, and ByteTrack multi-object tracking.
- Implemented confidence accumulation and persistent tracking logic to reduce false positives and preserve identity continuity across video frames.
- Supported webcam, video-file, and RTSP inputs for real-time monitoring scenarios.
- Containerized the application with **Docker** for reproducible deployment and environment setup.

AI Chatbot with Vision | *Llama 3 | OpenCV | Python | Multimodal AI*
- Built a vision-enabled chatbot that captures real-time image frames with **OpenCV** and passes visual context to **Llama 3** for conversational responses.
- Implemented the Python input/output pipeline needed to encode frames, communicate with the language model, and return responses to the user.

LifeGuard SOS Application | *Swift | iOS | Llama 3 | AI-Assisted Mobile App*
- Developed an iOS SOS application in **Swift** with Llama 3 integration, combining mobile emergency functionality with AI-assisted interaction.
- Demonstrated cross-platform engineering ability beyond Python by integrating AI functionality into a native mobile application.

Criminal Face Detection | *C++ | OpenCV | Face Recognition*
- Built an educational face-recognition application in **C++ and OpenCV** with a local enrollment and recognition workflow.
- Collected **10 grayscale 100x100 portrait samples** per enrolled identity, stored face data locally, and trained the recognition workflow from captured samples.

ENGINEERING STRENGTHS

- End-to-end ownership: dataset preparation, custom training pipelines, validation, inference, API serving, optimization, and deployment.
- Real-time and edge AI: Jetson Orin Nano deployment, webcam/video/RTSP processing, and performance-conscious computer-vision pipelines.
- Retrieval engineering: large-corpus FAISS indexing, embedding selection, chunking strategy, and trade-offs among **RAM, latency, and retrieval accuracy**.
- Technical leadership: client-facing communication, mentoring, code/work review, and coordination of large international project teams.

EDUCATION

Master of Science in Computer Science | Sejong University **2026 - Present**
Seoul, South Korea
Focus: Artificial Intelligence, Computer Vision, Machine Learning, and AI Systems

Bachelor of Computer Science | Sejong University **2022 - 2026**
Seoul, South Korea
GPA: 3.75 / 4.5
Relevant Coursework: Deep Learning, Computer Vision, Pattern Recognition, Database Systems, C++

RECOGNITION & COMMUNICATION

People's Choice Award - Capstone Design 2025 | Sejong University | Smart Aquaculture Assistant
AI Expo 2026: Contributed core AI and retrieval components to the Korean military meal-planning system demonstrated at the event.
English: IELTS 6.5; used for client meetings, technical documentation, presentations, and coordination of international teams.
Vietnamese: Native | **German:** B1 | **Korean:** Basic